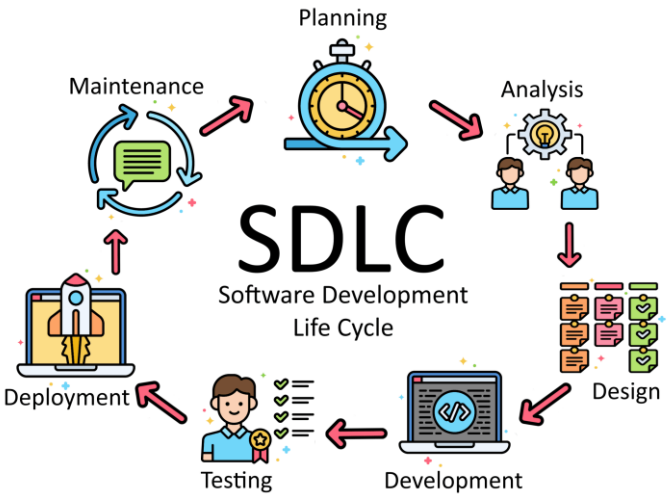
Software Development Life Cycle

SDLC (Software Development Life Cycle) is a structured process used to design, develop, test, and deploy software in a systematic and efficient way. It helps teams deliver high-quality software on time and within budget.

<u>Stages in SDLC:</u>



1. Planning:

In this stage, the project idea is defined and analyzed. Teams decide the scope, budget, timeline, and resources needed. Risks and feasibility are also evaluated.

Eg : A company decides to build an e-commerce website and plans budget, team size, and deadlines.

2. Requirement Analysis:

All business and user requirements are gathered and documented. This helps the team clearly understand what needs to be built. The final output is usually an SRS document.

Eg: Features like login, add to cart, payment gateway are defined

3. Design:

The system architecture and design are created based on requirements. It includes high-level and low-level design of components. UI, database, and system flow are planned here.

Eg : Designing how frontend connects to backend and database.

4. Development:

Developers start coding the application based on the design. Each module is built following coding standards. This is the longest phase of the SDLC.

Eg : Writing code for login page, APIs, database queries.

5. Testing:

The software is tested to find and fix bugs. Different testing methods ensure the application works as expected. It also verifies performance, security, and usability.

Eg: Checking if login works correctly or payment fails under errors.

6. Deployment:

The final product is released into the production environment. Users can now access and use the software. Deployment can be done all at once or in phases.

Eg: Launching the website for real customers.

7. Maintenance:

After deployment, the system is monitored and maintained. Bugs are fixed, and new features may be added. This ensures the software continues to perform well over time.

Eg: Adding new payment options or fixing user-reported issues